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Group Number

Intelligent Helmets for Motorcyclists

Group Members: *David Chase, Richard Osborne, William Ronan*

INTRODUCTION

The technological revolution of the information age has radically modernized nearly every industry and has changed the way humans live. A prime example of this can be found in the automotive industry where computer technology has completely reshaped driving; albeit the development of tech for motorcyclists has lagged [1]. The goal of this project is to help progress the development of computer aided safety features for motorcyclists by developing a prototype of the motorcycle helmet of the future.

To gain a better understanding of dangers that motorcyclists face, extensive research was done into the causes and outcomes of collisions [2][3][4]. Based on these findings, the primary objectives for this prototype are to provide:

- **Crash detection**
- **Emergency communication**
- **Object detection (front view and blind spots)**
- **Real-time GPS navigational features**
- **Integration with head's-up display**

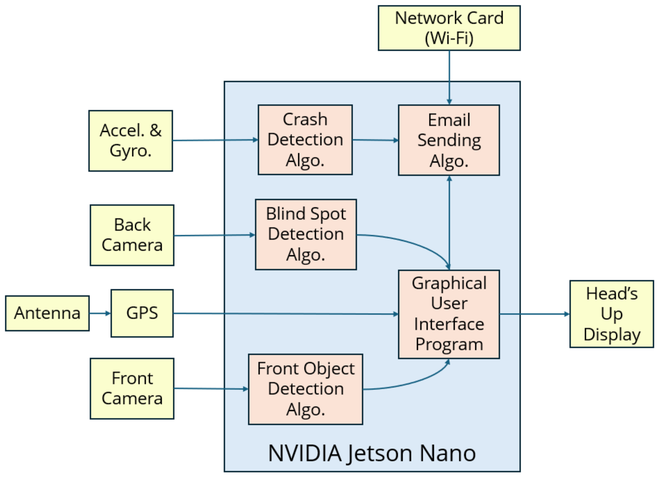
Emphasizing the need for real-time response, our modular approach prioritized runtime efficiency while remaining economically feasible, albeit with the trade-off in size and user interface.

ANALYSIS

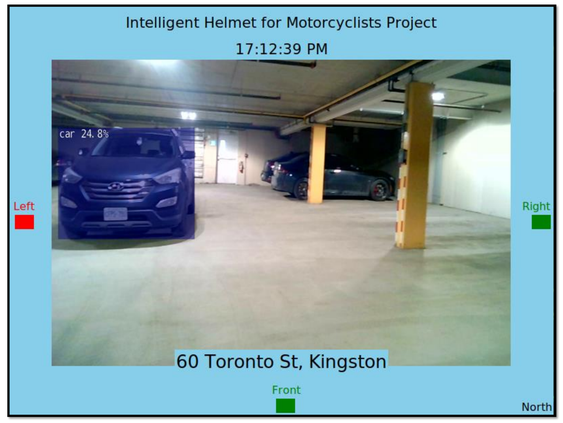
The subsystems with their respective analysis:

- **Crash detection:** Monitors accelerometer and gyroscope forces at 0.1-second intervals, assessing for jerk threshold breaches [5], and initiating emergency communication protocol accordingly
- **Emergency communication:** When a crash is detected, a TLS encrypted email is instantly sent to a user-defined contact, providing Google Maps location, GPS coordinates, and a photo of the scene
- **Object detection:** Trained on dash cam images of cars, pedestrians, road signs, and bicycles, the cameras and algorithm [6] operate at ~5 FPS, detecting objects in the user's blind spots and objects approaching from the front
- **Real-time GPS navigational features:** Translates longitude and latitude coordinates into easily understandable locations, and current directional heading information
- **Integration with head's up display:** Displays camera-captured images upon object detection, with symbols indicating left, right, or front detection; presents real-time GPS navigational features (address, directional heading) and the current time

TABLES / FIGURES / DIAGRAMS



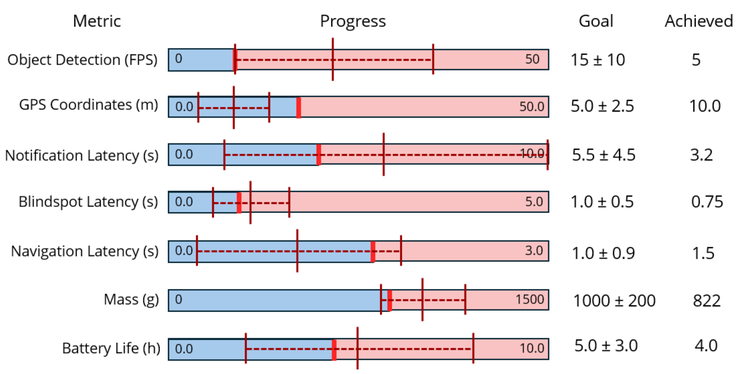
(a) Assembly of system components



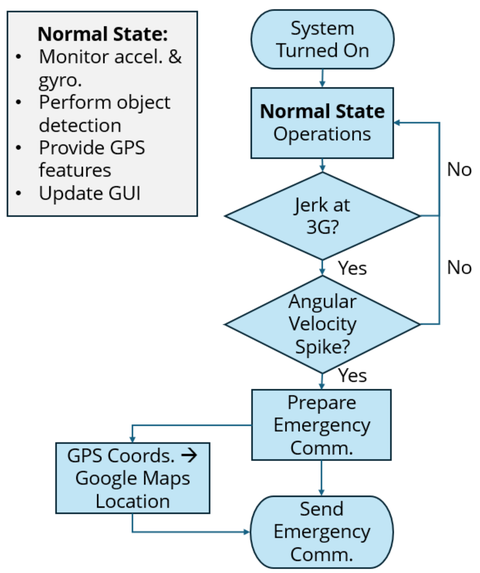
(b) View of head's up display during blind spot detection



(c) Sample camera-captures in various settings



(d) Progress bars for various project metrics: goal v. achieved



(e) Flow diagram for crash detection

CONCLUSIONS & RECOMMENDATIONS

The completed bench testing for individual components, integration testing for the whole system, and functional real-time environmental testing and validation of our prototype demonstrates that it is feasible and advantageous to improve the safety of a motorcyclist through a helmet-mounted intelligent device. **Major successes:**

- The object detection functions well in most environments of various lighting and weather conditions
- Accomplished all five primary objectives that were listed within the limited budget and timeline
- Navigated the project with a paramount focus on safety, effectively lowering the barrier to entry for future developments aimed at enhancing motorcyclist safety through artificial intelligence

On **future iterations**, with a larger budget:

- Facilitate efficient user interaction through implementation of a mobile application or integrating a touch-enabled LCD screen
- Reduce product size to minimize helmet weight and increase aerodynamics
- Implement cellular connectivity to enable practical (text, call) emergency communication
- Add the ability to scan and interpret the symbols and writing on road signs to be displayed on the GUI

REFERENCES

[1] J. Bright, "How much tech is too much tech on a motorcycle?," Revzilla, 19 February 2019. [Online]. Available: https://www.revzilla.com/common-tread/revisited-how-much-tech-is-too-much-tech-on-a-motorcycle. [Accessed 10 November 2023].
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